

Unit 25: Sigma Notation and Limits of Finite Sums — Full Solutions

Warm-up

$$1. \sum_{i=1}^5 i = 1 + 2 + 3 + 4 + 5 = 15$$

$$2. \sum_{i=1}^4 i^2 = 1 + 4 + 9 + 16 = 30$$

$$3. \sum_{i=1}^4 (2i+1): \text{ at } i = 1, 2, 3, 4: 3, 5, 7, 9.$$

$$3 + 5 + 7 + 9 = 24$$

$$4. \sum_{i=1}^3 i^3 = 1 + 8 + 27 = 36$$

$$5. \sum_{i=1}^6 3 = 3 \times 6 = 18$$

$$6. \sum_{i=1}^5 (i-1): \text{ at } i = 1, 2, 3, 4, 5: 0, 1, 2, 3, 4.$$

$$0 + 1 + 2 + 3 + 4 = 10$$

Standard

$$7. \sum_{i=1}^{10} i = \frac{10(11)}{2} = 55$$

$$8. \sum_{i=1}^{10} i^2 = \frac{10(11)(21)}{6} = \frac{2310}{6} = 385$$

$$9. \sum_{i=1}^6 i^3 = \left[\frac{6(7)}{2} \right]^2 = [21]^2 = 441$$

$$10. \sum_{i=1}^8 (3i-2)$$

$$= 3 \sum_{i=1}^8 i - \sum_{i=1}^8 2 = 3 \left(\frac{8(9)}{2} \right) - 2(8) = 3(36) - 16 = 108 - 16 = 92$$

$$11. \sum_{i=1}^5 (i^2 + 2i)$$

$$= \sum_{i=1}^5 i^2 + 2 \sum_{i=1}^5 i = \frac{5(6)(11)}{6} + 2 \left(\frac{5(6)}{2} \right) = \frac{330}{6} + 2(15) = 55 + 30 = 85$$

$$12. \sum_{i=1}^{20} 5 = 5 \times 20 = 100$$

Challenge

13. Exact area under $f(x) = x^2$ on $[0, 3]$.

$$\Delta x = \frac{3}{n}, \text{ and } x_i = \frac{3i}{n}.$$

$$f(x_i) \cdot \Delta x = \left(\frac{3i}{n} \right)^2 \cdot \frac{3}{n} = \frac{9i^2}{n^2} \cdot \frac{3}{n} = \frac{27i^2}{n^3}$$

$$R_n = \sum_{i=1}^n \frac{27i^2}{n^3} = \frac{27}{n^3} \sum_{i=1}^n i^2 = \frac{27}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{27(n+1)(2n+1)}{6n^2}$$

Expand $(n+1)(2n+1) = 2n^2 + 3n + 1$:

$$R_n = \frac{27(2n^2 + 3n + 1)}{6n^2} = \frac{27}{6} \left(2 + \frac{3}{n} + \frac{1}{n^2} \right) = 4.5 \left(2 + \frac{3}{n} + \frac{1}{n^2} \right)$$

Take the limit as $n \rightarrow \infty$ (the $\frac{3}{n}$ and $\frac{1}{n^2}$ terms vanish):

$$\lim_{n \rightarrow \infty} R_n = 4.5(2) = 9$$

Exact area = 9.

14. Exact area under $f(x) = x$ on $[0, 4]$.

$$\Delta x = \frac{4}{n}, \quad x_i = \frac{4i}{n}.$$

$$f(x_i) \Delta x = \frac{4i}{n} \cdot \frac{4}{n} = \frac{16i}{n^2}$$

$$R_n = \frac{16}{n^2} \sum_{i=1}^n i = \frac{16}{n^2} \cdot \frac{n(n+1)}{2} = \frac{8(n+1)}{n} = 8 + \frac{8}{n}$$

$$\lim_{n \rightarrow \infty} R_n = 8$$

Exact area = 8.

15. Exact area under $f(x) = 2x + 1$ on $[0, 2]$.

$$\Delta x = \frac{2}{n}, \quad x_i = \frac{2i}{n}.$$

$$f(x_i) = 2\left(\frac{2i}{n}\right) + 1 = \frac{4i}{n} + 1$$

$$f(x_i)\Delta x = \left(\frac{4i}{n} + 1\right) \cdot \frac{2}{n} = \frac{8i}{n^2} + \frac{2}{n}$$

$$\begin{aligned} R_n &= \frac{8}{n^2} \sum_{i=1}^n i + \frac{2}{n} \sum_{i=1}^n 1 = \frac{8}{n^2} \cdot \frac{n(n+1)}{2} + \frac{2}{n} \cdot n \\ &= \frac{4(n+1)}{n} + 2 = 4 + \frac{4}{n} + 2 = 6 + \frac{4}{n} \end{aligned}$$

$$\lim_{n \rightarrow \infty} R_n = 6$$

Exact area = 6.

16. Exact area under $f(x) = x^2$ on $[1, 2]$.

$$\Delta x = \frac{1}{n}, \quad x_i = 1 + \frac{i}{n}.$$

$$f(x_i) = \left(1 + \frac{i}{n}\right)^2 = 1 + \frac{2i}{n} + \frac{i^2}{n^2}$$

$$f(x_i)\Delta x = \left(1 + \frac{2i}{n} + \frac{i^2}{n^2}\right) \cdot \frac{1}{n} = \frac{1}{n} + \frac{2i}{n^2} + \frac{i^2}{n^3}$$

$$\begin{aligned} R_n &= \sum_{i=1}^n \frac{1}{n} + \frac{2}{n^2} \sum_{i=1}^n i + \frac{1}{n^3} \sum_{i=1}^n i^2 \\ &= n \cdot \frac{1}{n} + \frac{2}{n^2} \cdot \frac{n(n+1)}{2} + \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} \\ &= 1 + \frac{n+1}{n} + \frac{(n+1)(2n+1)}{6n^2} \end{aligned}$$

Simplify each piece: $\frac{n+1}{n} = 1 + \frac{1}{n}$. And $\frac{(n+1)(2n+1)}{6n^2} = \frac{2n^2+3n+1}{6n^2} = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}$.

$$R_n = 1 + \left(1 + \frac{1}{n}\right) + \left(\frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}\right) = \frac{7}{3} + \frac{3}{2n} + \frac{1}{6n^2}$$

$$\lim_{n \rightarrow \infty} R_n = \frac{7}{3}$$

$$\text{Exact area} = \frac{7}{3}.$$

$$17. \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \left(\frac{i}{n}\right)^2$$

Rewrite the summand:

$$\frac{1}{n} \left(\frac{i}{n}\right)^2 = \frac{i^2}{n^3}$$

$$\frac{1}{n} \sum_{i=1}^n \left(\frac{i}{n}\right)^2 = \frac{1}{n^3} \sum_{i=1}^n i^2 = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6n^2}$$

$$\text{Expand: } \frac{2n^2 + 3n + 1}{6n^2} = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}.$$

$$\lim_{n \rightarrow \infty} \left(\frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2} \right) = \frac{1}{3}$$

Answer: $\frac{1}{3}$. (This is secretly the exact area under $f(x) = x^2$ on $[0, 1]$, written entirely in limit-of-sums form.)

Applied

$$18. v(t) = 2t \text{ on } [0, 5].$$

$$\Delta t = \frac{5}{n}, t_i = \frac{5i}{n}.$$

$$v(t_i) \Delta t = 2 \left(\frac{5i}{n}\right) \cdot \frac{5}{n} = \frac{50i}{n^2}$$

$$R_n = \frac{50}{n^2} \sum_{i=1}^n i = \frac{50}{n^2} \cdot \frac{n(n+1)}{2} = \frac{25(n+1)}{n} = 25 + \frac{25}{n}$$

$$\lim_{n \rightarrow \infty} R_n = 25$$

Answer: exact total distance = 25 ft.

$$19. r(t) = t^2 \text{ on } [0, 4].$$

$$\Delta t = \frac{4}{n}, t_i = \frac{4i}{n}.$$

$$r(t_i) \Delta t = \left(\frac{4i}{n}\right)^2 \cdot \frac{4}{n} = \frac{16i^2}{n^2} \cdot \frac{4}{n} = \frac{64i^2}{n^3}$$

$$R_n = \frac{64}{n^3} \sum_{i=1}^n i^2 = \frac{64}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{64(n+1)(2n+1)}{6n^2}$$

Expand: $\frac{64(2n^2+3n+1)}{6n^2} = \frac{32}{3} \left(2 + \frac{3}{n} + \frac{1}{n^2} \right).$

$$\lim_{n \rightarrow \infty} R_n = \frac{32}{3}(2) = \frac{64}{3}$$

Answer: exact total water collected = $\frac{64}{3}$ gallons.

20. $MR(x) = 6x$ on $[0, 10]$.

$$\Delta x = \frac{10}{n}, \quad x_i = \frac{10i}{n}.$$

$$MR(x_i)\Delta x = 6 \left(\frac{10i}{n} \right) \cdot \frac{10}{n} = \frac{600i}{n^2}$$

$$R_n = \frac{600}{n^2} \sum_{i=1}^n i = \frac{600}{n^2} \cdot \frac{n(n+1)}{2} = \frac{300(n+1)}{n} = 300 + \frac{300}{n}$$

$$\lim_{n \rightarrow \infty} R_n = 300$$

Answer: exact total revenue = \$300.

21. $v(t) = 3t^2$ on $[0, 4]$.

$$\Delta t = \frac{4}{n}, \quad t_i = \frac{4i}{n}.$$

$$v(t_i)\Delta t = 3 \left(\frac{4i}{n} \right)^2 \cdot \frac{4}{n} = 3 \cdot \frac{16i^2}{n^2} \cdot \frac{4}{n} = \frac{192i^2}{n^3}$$

$$R_n = \frac{192}{n^3} \sum_{i=1}^n i^2 = \frac{192}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{32(n+1)(2n+1)}{n^2}$$

Expand: $\frac{32(2n^2+3n+1)}{n^2} = 32 \left(2 + \frac{3}{n} + \frac{1}{n^2} \right) = 64 + \frac{96}{n} + \frac{32}{n^2}.$

$$\lim_{n \rightarrow \infty} R_n = 64$$

Answer: exact total distance traveled = 64 ft.